Data from a validation study of two psychometric models on test-taking behavior

CODEBOOK

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Naming conventions for measurement instruments

Prefix, root and suffix are separated by an underscore (_)

Prefix

X – Raw response
Y – Scored response
T – Response time
CR – Confidence rating
No prefix – Raw response is the same as scored (X = Y) and this is what the variable contains

Root

Instrument abbreviation + Item number (with no underscore)
No item number – The variable is total score

MR – Matrix Reasoning
MRt – Matrix Reasoning training
MRm – Matrix Reasoning masking
MRp – Unsolvable persistence items
CT – Concentration Task
CTt – Concentration Task Training
CM – Current Motivation
EF – Effort
AO – Action Orientation (Motivational Disposition)

Suffix

Block/Timepoint number
No suffix – instrument is administered only at one timepoint

Missing data

In the dataset, missing data are represented with an empty entry. There is no specific numeric or other value associated with missing data.

Sociodemographic variables

Variable	Description	Type	Values/Units
Age	Age of the participant	Numeric	Years
Gender	Gender of the participant	Character	Male, Female, Other
Education	Highest degree of education obtained by participant	Factor (character)	Education level
Language	Native language of the participant	Character	Language name (multiple languages are comma-separated)
Country	Residence country of the participant	Character	Country name

Study variables

Variable	Description	Type	Values/Units
ID	Participant ID	Character	Numeric string
Group	Study group	Character	1 (group which first had non-speeded instruction), 2 (group which first had speeded instructions)
AC	Number of passed attention checks	Factor (numeric)	0-3

Usage variables

Variable	Description	Type	Values/Units
Device	Device from which the study was accessed	Character	Computer, Tablet
Disruptions	Disruptions experienced by participant during the session	Character	Character vector containing one of 11 disruption categories

Matrix Reasoning Test

Variable	Description	Type	Values/Units
X_MR01_01	Raw response to the first item	Character vector	An array of 0s and 1s indicating whether the element was selected[1]
X_MR02_01	Raw response to the second item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR03_01	Raw response to the third item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR04_01	Raw response to the fourth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR05_01	Raw response to the fifth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR06_01	Raw response to the sixth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR07_01	Raw response to the seventh item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR08_01	Raw response to the eighth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR09_01	Raw response to the ninth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR10_01	Raw response to the tenth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR11_02	Raw response to the eleventh item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR12_02	Raw response to the twelveth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR13_02	Raw response to the thirteenth item	Character vector	An array of 0s and 1s indicating whether the element was selected

[1] The matrix reasoning task is composed of a set of squares showing certain patterns, with the last square missing. The pattern in the last square has to be constructed from 20 elements given to the participant. The vector represents each element participant selected with “1” and each element participant didn’t select with “0”.

Matrix Reasoning Test

Variable	Description	Type	Values/Units
X_MR14_02	Raw response to the fourteenth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR15_02	Raw response to the fifteenth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR16_02	Raw response to the sixteenth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR17_02	Raw response to the seventeenth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR18_02	Raw response to the eighteenth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR19_02	Raw response to the nineteenth item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MR20_02	Raw response to the twentieth item	Character vector	An array of 0s and 1s indicating whether the element was selected
Y_MR01_01	Scored response to the first item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR02_01	Scored response to the second item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR03_01	Scored response to the third item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR04_01	Scored response to the fourth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR05_01	Scored response to the fifth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR06_01	Scored response to the sixth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR07_01	Scored response to the seventh item	Factor (numeric)	0 (incorrect response) or 1 (correct response)

Matrix Reasoning Test

Variable	Description	Type	Values/Units
Y_MR08_01	Scored response to the eighth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR09_01	Scored response to the ninth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR10_01	Scored response to the tenth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR11_02	Scored response to the eleventh item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR12_02	Scored response to the twelveth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR13_02	Scored response to the thirteenth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR14_02	Scored response to the fourteenth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR15_02	Scored response to the fifteenth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR16_02	Scored response to the sixteenth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR17_02	Scored response to the seventeenth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR18_02	Scored response to the eighteenth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR19_02	Scored response to the nineteenth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MR20_02	Scored response to the twentieth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
T_MR01_01	Response time on the first item	Numeric	Milliseconds
T_MR02_01	Response time on the second item	Numeric	Milliseconds

Matrix Reasoning Test

Variable	Description	Type	Values/Units
T_MR03_01	Response time on the third item	Numeric	Milliseconds
T_MR04_01	Response time on the fourth item	Numeric	Milliseconds
T_MR05_01	Response time on the fifth item	Numeric	Milliseconds
T_MR06_01	Response time on the sixth item	Numeric	Milliseconds
T_MR07_01	Response time on the seventh item	Numeric	Milliseconds
T_MR08_01	Response time on the eighth item	Numeric	Milliseconds
T_MR09_01	Response time on the ninth item	Numeric	Milliseconds
T_MR10_01	Response time on the tenth item	Numeric	Milliseconds
T_MR11_02	Response time on the eleventh item	Numeric	Milliseconds
T_MR12_02	Response time on the twelveth item	Numeric	Milliseconds
T_MR13_02	Response time on the thirteenth item	Numeric	Milliseconds
T_MR14_02	Response time on the fourteenth item	Numeric	Milliseconds
T_MR15_02	Response time on the fifteenth item	Numeric	Milliseconds
T_MR16_02	Response time on the sixteenth item	Numeric	Milliseconds
T_MR17_02	Response time on the seventeenth item	Numeric	Milliseconds
T_MR18_02	Response time on the eighteenth item	Numeric	Milliseconds

Matrix Reasoning Test

Variable	Description	Type	Values/Units
T_MR19_02	Response time on the nineteenth item	Numeric	Milliseconds
T_MR20_02	Response time on the twentieth item	Numeric	Milliseconds
Y_MRt01	Scored response on the first training item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MRt02	Scored response on the second training item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MRt03	Scored response on the third training item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MRt04	Scored response on the fourth training item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MRt05	Scored response on the fitth training item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_MRt06	Scored response on the sixth training item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
X_MRt01	Raw response on the first training item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MRt02	Raw response on the second training item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MRt03	Raw response on the third training item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MRt04	Raw response on the fourth training item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MRt05	Raw response on the fitth training item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MRt06	Raw response on the sixth training item	Character vector	An array of 0s and 1s indicating whether the element was selected

Matrix Reasoning Test

Variable	Description	Type	Values/Units
T_MRt01	Response time to the first training item	Numeric	Milliseconds
T_MRt02	Response time to the second training item	Numeric	Milliseconds
T_MRt03	Response time to the third training item	Numeric	Milliseconds
T_MRt04	Response time to the fourth training item	Numeric	Milliseconds
T_MRt05	Response time to the fitth training item	Numeric	Milliseconds
T_MRt06	Response time to the sixth training item	Numeric	Milliseconds
Y_MRm01	Scored response on the masking item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
X_MRm01	Raw response on the masking item	Character vector	An array of 0s and 1s indicating whether the element was selected
T_MRm01	Response time on the masking item	Numeric	Milliseconds
Y_MR_01	Total score in the first block	Numeric	Number of correct items
Y_MR_02	Total score in the second block	Numeric	Number of correct items
Y_MRt	Total score on the training items	Numeric	Number of correct items

Confidence ratings

Variable	Description	Type	Values
CR_MRt01	Confidence rating to the correctness of the response to the first item in the matrix test training set	Numeric	0-100
CR_MRt02	Confidence rating to the correctness of the response to the second item in the matrix test training set	Numeric	0-100
CR_MRt03	Confidence rating to the correctness of the response to the third item in the matrix test training set	Numeric	0-100
CR_MRt04	Confidence rating to the correctness of the response to the fourth item in the matrix test training set	Numeric	0-100
CR_MRt05	Confidence rating to the correctness of the response to the fifth item in the matrix test training set	Numeric	0-100
CR_MRt06	Confidence rating to the correctness of the response to the sixth item in the matrix test training set	Numeric	0-100

Concentration task			
Variable	**Description**	**Type**	**Values/Units**
X_CT01_01	Raw response to the first item	Numeric	Number of counted target letters
X_CT02_01	Raw response to the second item	Numeric	Number of counted target letters
X_CT03_01	Raw response to the third item	Numeric	Number of counted target letters
X_CT04_02	Raw response to the fourth item	Numeric	Number of counted target letters
X_CT05_02	Raw response to the fifth item	Numeric	Number of counted target letters
X_CT06_02	Raw response to the sixth item	Numeric	Number of counted target letters
X_CT07_03	Raw response to the seventh item	Numeric	Number of counted target letters
X_CT08_03	Raw response to the eighth item	Numeric	Number of counted target letters
X_CT09_03	Raw response to the ninth item	Numeric	Number of counted target letters
Y_CT01_01	Scored response to the first item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_CT02_01	Scored response to the second item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_CT03_01	Scored response to the third item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_CT04_02	Scored response to the fourth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_CT05_02	Scored response to the fifth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_CT06_02	Scored response to the sixth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_CT07_03	Scored response to the seventh item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_CT08_03	Scored response to the eighth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)

Concentration task			
Variable	**Description**	**Type**	**Values/Units**
Y_CT09_03	Scored response to the ninth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
X_CTt	Raw response to the training item	Numeric	Number of counted target letters
Y_CTt	Scored response to the training item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
Y_CT_01	Total score at the first measurement point	Numeric	Sum of scored responses in the first measurement point
Y_CT_02	Total score at the second measurement point	Numeric	Sum of scored responses in the second measurement point
Y_CT_03	Total score at the third measurement point	Numeric	Sum of scored responses in the third measurement point

Current motivation

Variable	Description	Type	Values/Units
CM01_01	Response to the first item at first measurement point	Numeric	0-100
CM02_01	Response to the second item at first measurement point	Numeric	0-100
CM03_01	Response to the third item at first measurement point	Numeric	0-100
CM01_02	Response to the first item at second measurement point	Numeric	0-100
CM02_02	Response to the second item at second measurement point	Numeric	0-100
CM03_02	Response to the third item at second measurement point	Numeric	0-100
CM01_03	Response to the first item at third measurement point	Numeric	0-100
CM02_03	Response to the second item at third measurement point	Numeric	0-100
CM03_03	Response to the third item at third measurement point	Numeric	0-100
CM_01	Total score at the first measurement point	Numeric	Mean of item responses
CM_02	Total score at the first measurement point	Numeric	Mean of item responses
CM_03	Total score at the first measurement point	Numeric	Mean of item responses

Effort

Variable	Description	Type	Values/Units
EF01_01	Response to the first item at first measurement point	Factor (numeric)	1-4
EF02_01	Response to the second item at first measurement point	Factor (numeric)	1-4
EF03_01	Response to the third item at first measurement point	Factor (numeric)	1-4
EF04_01	Response to the fourth item at first measurement point	Factor (numeric)	1-4
EF05_01	Response to the fifth item at first measurement point	Factor (numeric)	1-4
EF01_02	Response to the first item at second measurement point	Factor (numeric)	1-4
EF02_02	Response to the second item at second measurement point	Factor (numeric)	1-4
EF03_02	Response to the third item at second measurement point	Factor (numeric)	1-4
EF04_02	Response to the fourth item at second measurement point	Factor (numeric)	1-4
EF05_02	Response to the fifth item at second measurement point	Factor (numeric)	1-4
EF_01	Total score at first measurement point	Numeric	Sum of item responses
EF_02	Total score at second measurement point	Numeric	Sum of item responses

Motivational disposition

Variable	Description	Type	Values
AO01	Response to the first item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response[2])
AO02	Response to the second item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO03	Response to the third item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO04	Response to the fourth item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO05	Response to the fifth item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO06	Response to the sixth item	Factor (numeric)	0 (incorrect response) or 1 (correct response)
AO07	Response to the seventh item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO08	Response to the eighth item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO09	Response to the ninth item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO10	Response to the tenth item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO11	Response to the eleventh item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response)
AO12	Response to the twelveth item	Factor (numeric)	0 (nonindicative response) or 1 (indicative response))
AO	Total score	Numeric	Number of indicative responses

[2] In personality and attitude measurement, an indicative response is a response which endorses the statements given in the item.

Persistence

Variable	Description	Type	Values
X_MRp01	Raw response (vector indicating selected elements) to the first persistence item	Character vector	An array of 0s and 1s indicating whether the element was selected
X_MRp02	Raw response (vector indicating selected elements) to the second persistence item	Character vector	An array of 0s and 1s indicating whether the element was selected
T_MRp01	Response time to the first persistence item	Numeric	Milliseconds
T_MRp02	Response time to the second persistence item	Numeric	Milliseconds